

Title

AI-Based Image Colorization: Deep Learning Improvement on Black & White Photographs

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1. Introduction

The proposed generative AI model aims to develop a state-of-the-art image colorization system for black-and-white photographs. Using deep learning, this model will focus on automatically generating realistic colorizations of monochromatic images, preserving fine details and ensuring vivid, accurate colors.

2. Problem Definition

The specific problem is the restoration and enhancement of black-and-white images by accurately predicting and applying colors. This task has significant applications in digital art restoration, historical photograph restoration, and film colorization.

Significance and Impact

Colorization of black-and-white images brings historical photographs to life, making them more engaging and accessible to modern audiences. It also has commercial applications in content creation and restoration industries. Successful implementation of this model will provide an efficient and scalable solution for image restoration, enhancing cultural preservation and storytelling through visual media.

3. Dataset Description

Dataset Details:Dataset Source:Kaggle

1. **Size and Format:** The dataset contains over 800 high-resolution images, stored in PNG format.
2. **Data Types:** The dataset comprises photographic images of various scenes, objects, and landscapes, including people, animals, historic monuments, and landscapes.
3. **Preprocessing:**
 - a. **Cleaning:** Ensure all images are valid and non-corrupt.
 - b. **Normalization:** Scale pixel values to a range of 0-1 to improve model convergence.
 - c. **Augmentation:** Apply random transformations such as rotations and flips to increase data diversity and improve generalization.

4. Problem-Data Alignment

The DIV2K dataset aligns well with the image colorization problem, as it contains diverse and high-resolution photographic images that can be converted to grayscale and then recolored by the model.

Suitability

The dataset is suitable due to its size, diversity, and high resolution, which will help train a generative AI model to capture intricate details and complex color distributions accurately.

5. Generative Model Description

This problem will be solved using a Pix2Pix GAN that consists of a generator trained to map grayscale images to colorful images and a discriminator trained to assess the realism of generated images.

Model Justification

Pix2Pix GANs are chosen because they are specifically designed for paired image-to-image translation tasks. This makes them ideal for learning the mapping from grayscale to colorized images. Additionally,

they can generate high-quality, realistic outputs by leveraging adversarial training to refine details and color accuracy.

6. Evaluation Metrics

The performance of the colorization model will be evaluated through both qualitative and quantitative metrics to ensure realism and accuracy.

Metrics

1. **Peak Signal-to-Noise Ratio (PSNR):** Measures the similarity between the colorized and ground truth images, with higher values indicating better quality.
2. **Structural Similarity Index (SSIM):** Evaluates the perceptual quality of the colorized images.
3. **Frechet Inception Distance (FID):** Measures the distributional distance between the generated and real images to assess realism.

7. Ethical Considerations

1. **Bias and Representation:** The model might produce biased colorizations if the training dataset is not diverse.
2. **Misinformation:** There is a risk of creating misleading historical representations by applying incorrect or unrealistic colors.

8. References

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